Exploratory Factor Analysis

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Introduction

Exploratory factor analysis (EFA) searches for a small number of latent factors that explain the correlations among a larger set of observed variables. It is the cornerstone of psychometric scale development, the standard tool for validating questionnaire structures, and a useful exploratory technique in any context where many correlated indicators are believed to reflect a smaller number of latent constructs. The output — rotated factor loadings together with factor correlations — directly informs item-selection, scale-construction, and CFA model-specification decisions.

Prerequisites

A working understanding of correlation matrices, principal components analysis, and the conceptual difference between observed indicators and latent constructs.

Theory

The common factor model is

$$X = \Lambda F + U,$$

where X is the observed item matrix, Λ is the loading matrix, F are the common factors, and U is unique (specific) variance. Unlike PCA, which lumps common and specific variance together, EFA explicitly partitions the variance.

The standard EFA workflow has five steps:

1. **Check factorability**: Kaiser-Meyer-Olkin (KMO > 0.6 acceptable, > 0.8 good) and Bartlett's sphericity test confirm that the correlation matrix has enough structure to factor.
2. **Choose number of factors**: parallel analysis is the modern gold standard; the obsolete Kaiser eigenvalue-greater-than-1 rule and the subjective scree plot are still seen but should be supplemented.
3. **Extract**: maximum likelihood (ML) for inferential purposes, principal-axis factoring (PAF) for robustness, principal-components for quick exploration.
4. **Rotate**: varimax (orthogonal) or oblimin / promax (oblique) for interpretable loadings.
5. **Interpret**: examine rotated loadings, factor correlations, and item communalities.

Assumptions

Multivariate Normal continuous data for ML extraction; linearity of item-factor relationships; sufficient sample size (rule of thumb: at least 5–10 cases per item, with $n \geq 200$ as a floor).

R Implementation

```r
library(psych); library(GPArotation)

d <- bfi[, 1:25]    # 25-item big-five inventory
```

```r
# Factorability
KMO(d)
cortest.bartlett(cor(d, use = "pairwise"), n = nrow(d))

# Number of factors
fa.parallel(d, fa = "fa")

# EFA with 5 factors, oblique rotation
efa <- fa(d, nfactors = 5, rotate = "oblimin", fm = "ml")
print(efa, cut = 0.3)
```

Output & Results

`KMO()` returns overall and per-item factorability indices; `cortest.bartlett()` returns the sphericity test. `fa.parallel()` overlays observed eigenvalues on simulated null eigenvalues to determine the number of factors. `fa()` returns rotated loadings, communalities, factor correlations (under oblique rotation), and explained variance per factor.

Interpretation

A reporting sentence: "EFA on 25 personality items (n = 2,800; KMO = 0.85, Bartlett's χ^2 = 18,245, $p < 0.001$) supported a five-factor solution by parallel analysis; oblimin-rotated loadings aligned with the Big Five structure (extraversion, neuroticism, conscientiousness, agreeableness, openness), explaining 45 % of common variance, and factor correlations ranged from 0.15 to 0.42." Always report KMO, Bartlett, the number-of-factors method, the extraction method, and the rotation.

Practical Tips

- Use parallel analysis (`fa.parallel`) for the number-of-factors decision; avoid the obsolete "eigenvalue > 1" Kaiser rule which over-extracts in most settings.
- Prefer oblique rotation (oblimin, promax) by default; uncorrelated factors are rare in real psychological or biological data.
- Watch for Heywood cases — loadings above 1 or communalities above 1 — which indicate over-extraction or model misspecification.
- For ordinal Likert items, use polychoric correlations (`fa(..., cor = "poly")`); Pearson correlations on Likert items underestimate the true factor structure.
- For confirmatory tests of the resulting structure, run a separate CFA (`lavaan::cfa`) on a held-out sample; never confirm on the data used to extract.
- Report item communalities alongside loadings; communalities below 0.20 indicate items that contribute little to any factor and may need removal.

R Packages Used

`psych` for `fa()`, `KMO()`, `fa.parallel()`, and the comprehensive EFA workflow; `GPArotation` for the underlying rotation algorithms; `EFAtools` for parallel-analysis-aware extraction with bootstrap confidence intervals; `lavaan` for confirmatory factor analysis as the natural follow-up; `MBESS` for additional reliability and structural-equation utilities.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**

- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- PCA, factor analysis, CCA, LDA
- Clustering: k-means, hierarchical, model-based
- UMAP and t-SNE